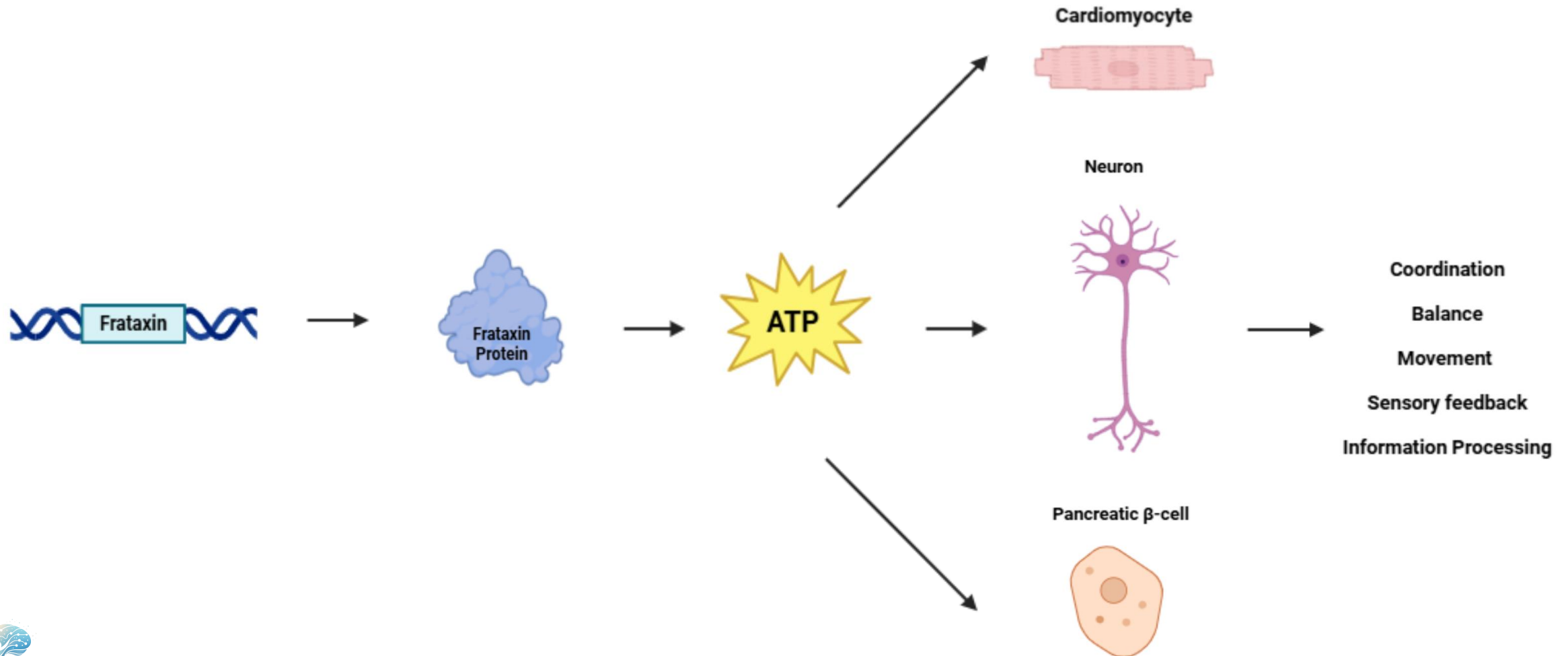




AtaxAI: Combating Neurodegeneration with Artificial Intelligence

Abby Micallef, Tamar Shalev, Rasik Sharma, Prabir Singh, and Dyan Socransky

Friedreich's ataxia results from a gene mutation, reducing frataxin and ATP production, leading to impaired neuronal function and various symptoms



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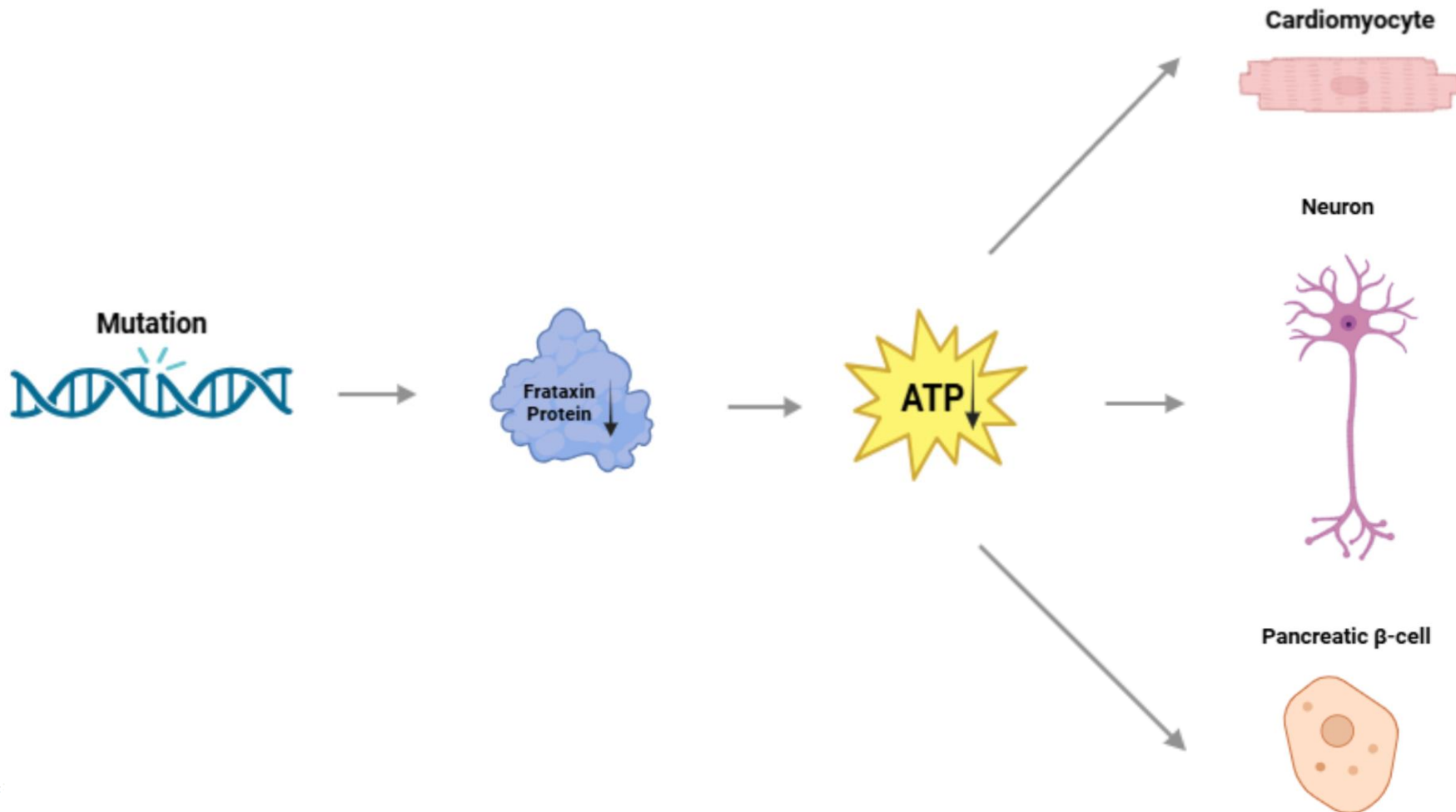
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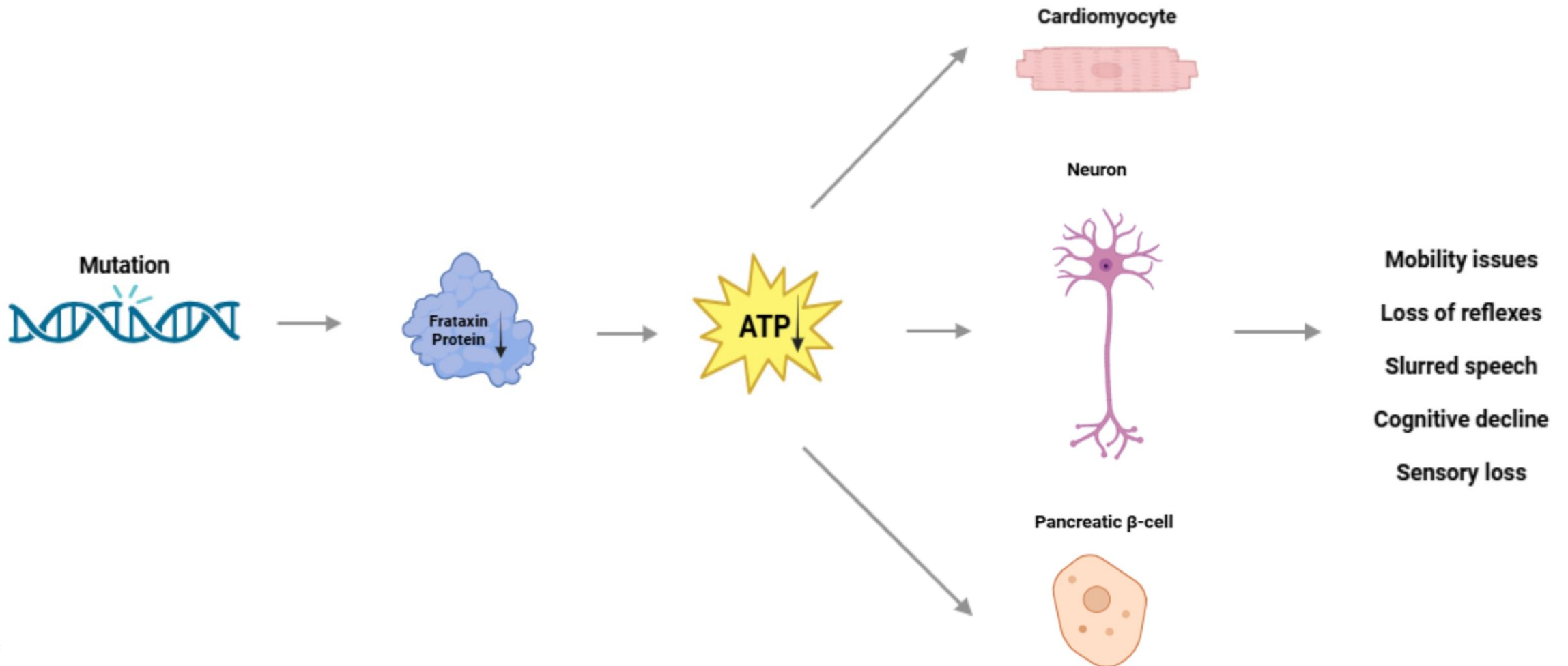
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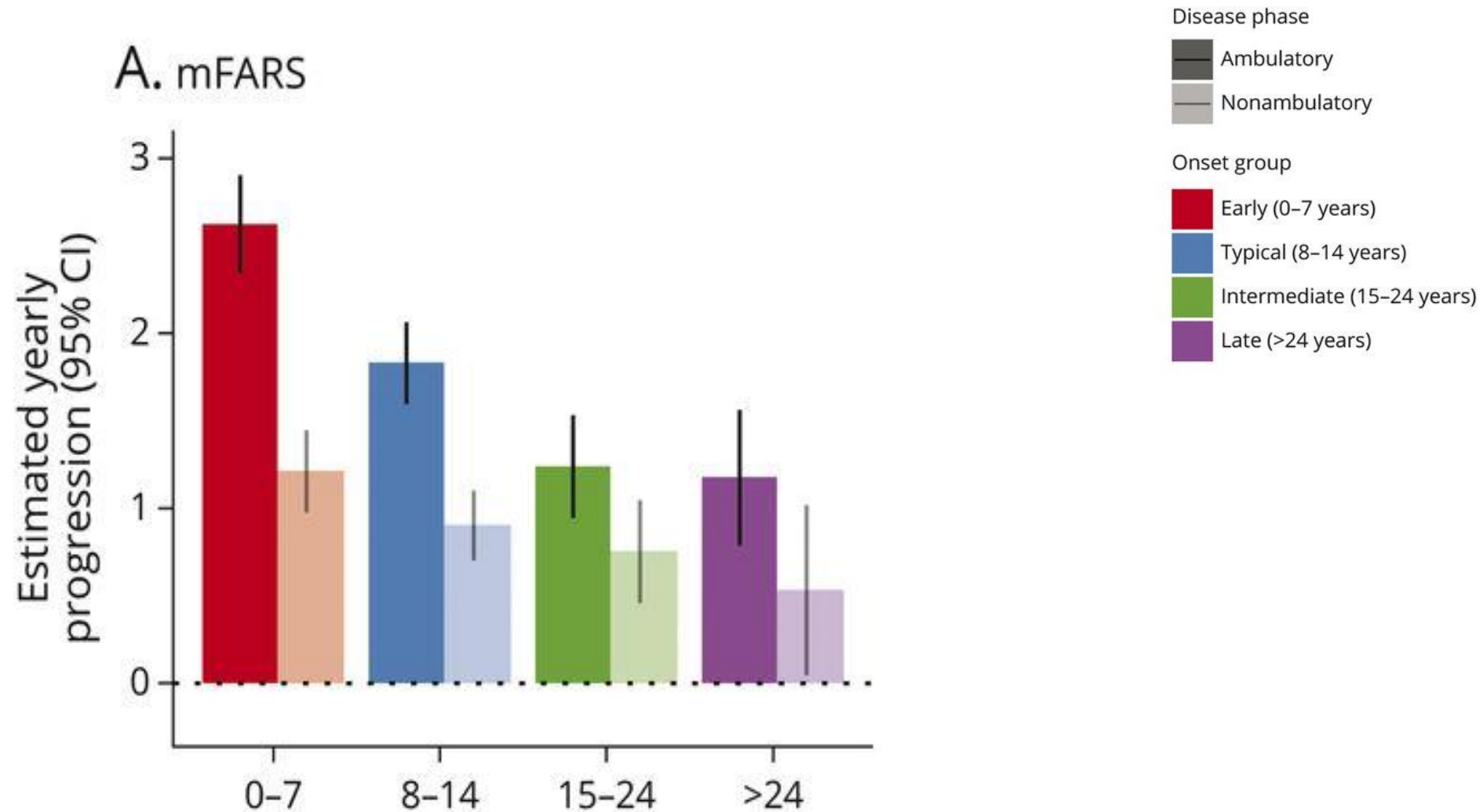
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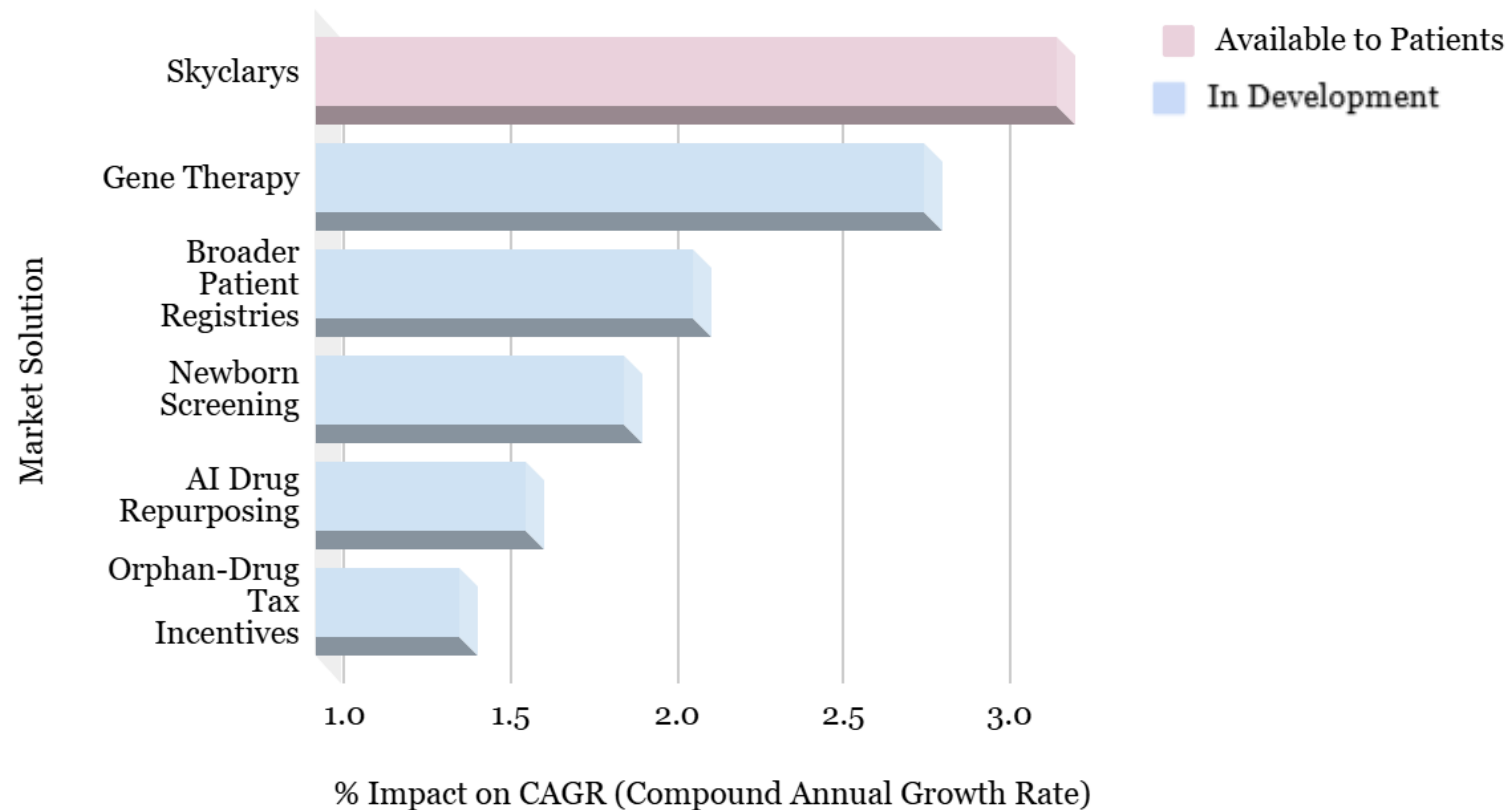


Friedreich's ataxia manifests uniquely in each patient, with faster progression in early onset cases

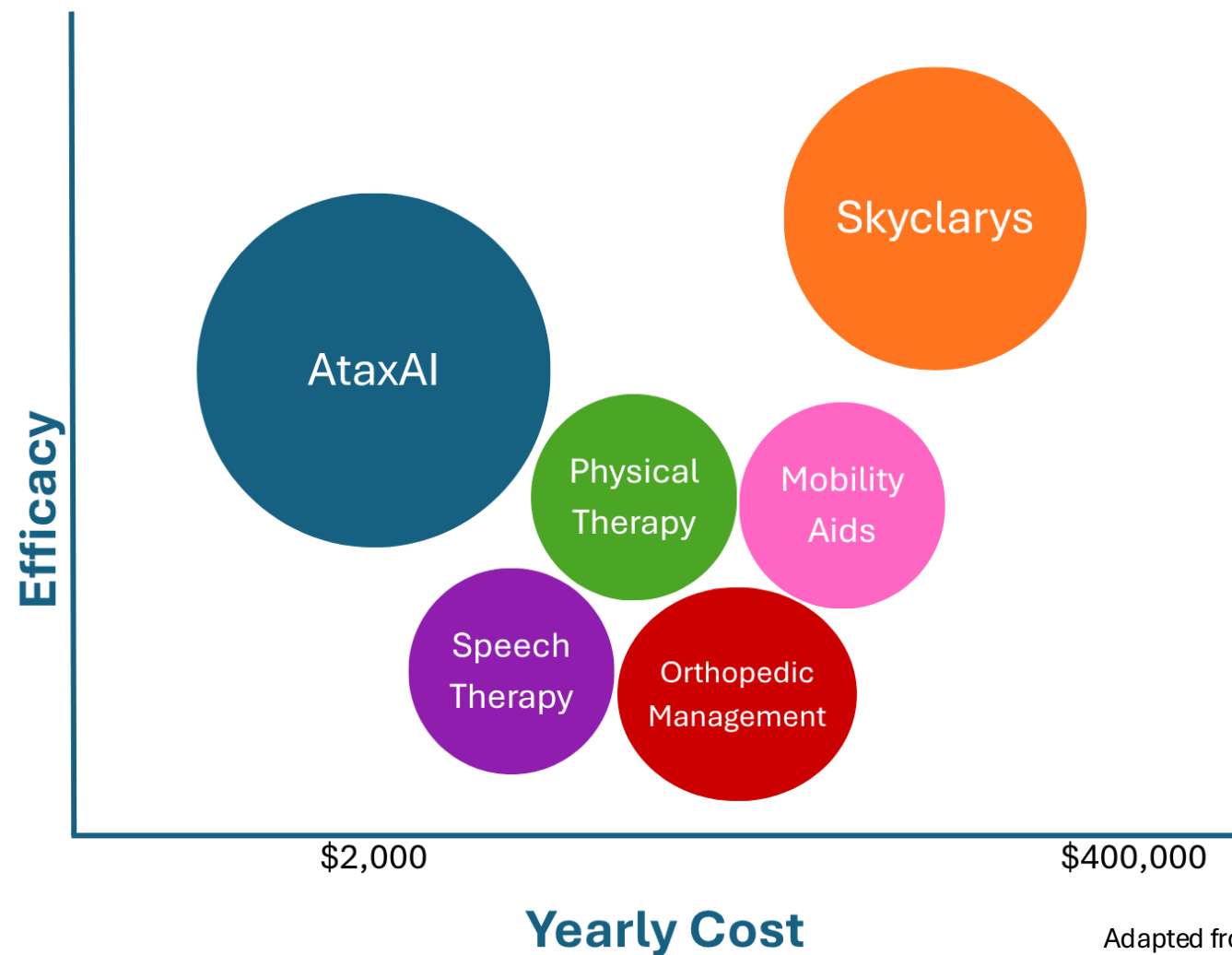


The current Friedreich's ataxia market of \$1.2 billion USD includes one approved drug, Skyclarys, and several treatments currently in trials

Current Friedreich's Ataxia Market



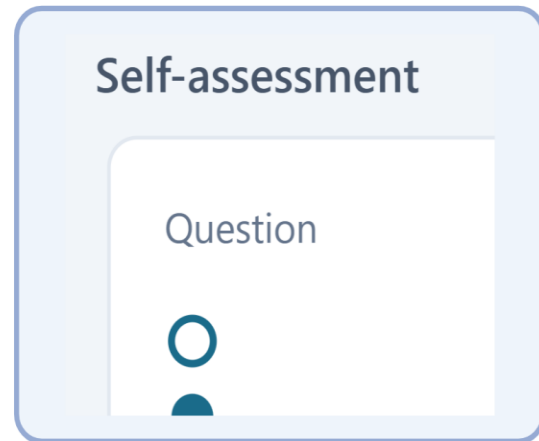
With limited available solutions, there is a gap in the market for Friedreich's ataxia treatments, specifically one that is medium efficacy and low-cost



Introducing: AtaxAI; an AI-powered cognitive & physical trainer based on patient-specific data



Medical Brain Imaging Data



Physiocognitive Self-Assessments



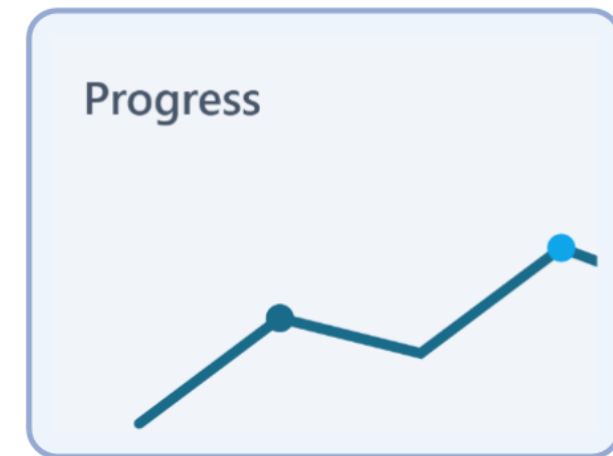
Patients can access AtaxAI consistently; daily use supports growth and targets personalized needs



Customized Enrichment Tasks



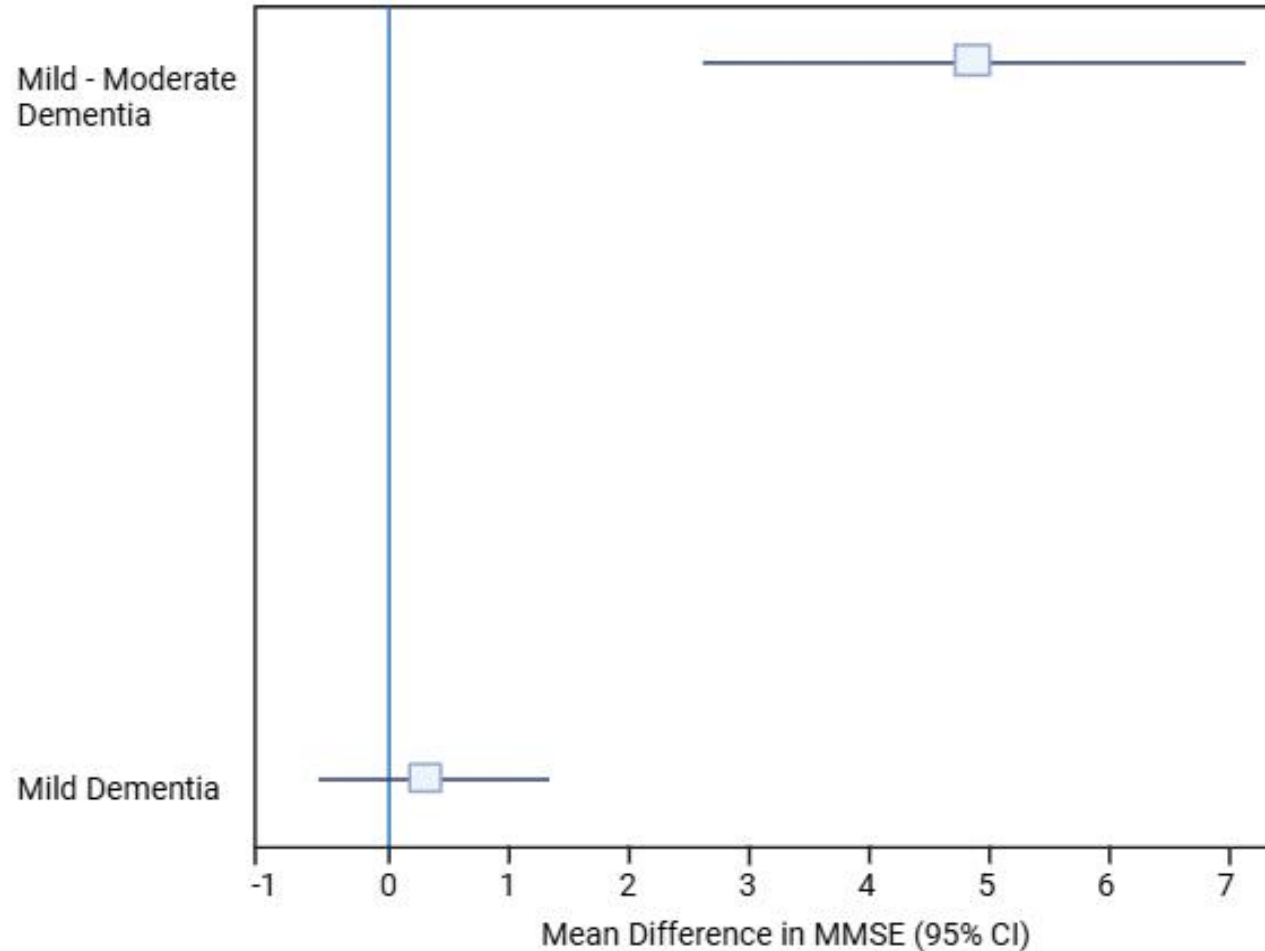
Real-time Progress Reports



Cognitive training shows a clear benefit in patients with mild-moderate dementia on Mini-Mental State Examination scale

Mild Dementia:
0.38 MMSE,
95%CI (-0.66 to
1.41)

Mild-Moderate
Dementia:
4.96MMSE,
95%CI (2.61 to
7.30)



Physical rehabilitation significantly improves motor performance and ataxia symptoms in patients with FRDA – with significant reduction on the FARS and SARA scales

	Pre (N = 42)	Post (N = 42)	Value of p^*
SARA	17.4 (7.8)	16.1 (7.9)	<0.001
FARS total	53.7 (17.2)	49.4 (17.7)	<0.001
<i>FARS LL</i>	<i>8.7 (4.4)</i>	<i>7.6 (4.5)</i>	<i><0.001</i>
<i>FARS UL</i>	<i>11.5 (4.6)</i>	<i>9.4 (4.2)</i>	<i><0.001</i>
<i>FARS US</i>	<i>18.5 (6.0)</i>	<i>17.4 (6.7)</i>	<i><0.001</i>

FARS – Friedrich's Ataxia Rating Scale (Lower Limb, Upper Limb, Upright Stability)

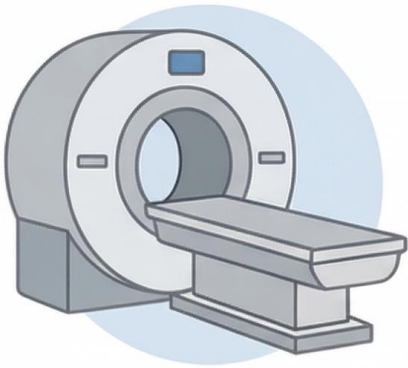
SARA – Scale for the Assessment and Rating of Ataxia

SARA and FARS scale measurements before and after physical rehabilitation.

Compared to other competitors, AtaxAI fulfills many conditions and requirements, making it a more versatile solution

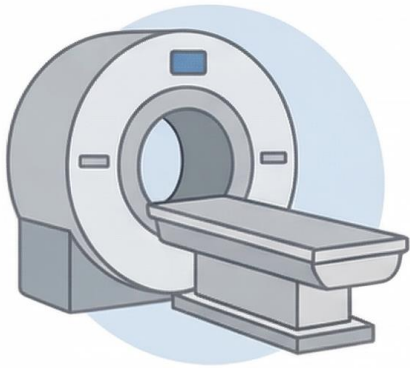
	Affordability	Personalization	Accessibility	Cognitive Outcomes	Physical Outcomes
Physiotherapy/ Rehab		✓	✓		✓
Skyclarys				✓	✓
Gene Therapy				✓	✓
AtaxAI	✓	✓	✓	✓	✓

Distribution of AtaxAI to patients through healthcare providers



1. CLINICAL ASSESSMENT

Distribution of AtaxAI to patients through healthcare providers



**1. CLINICAL
ASSESSMENT**

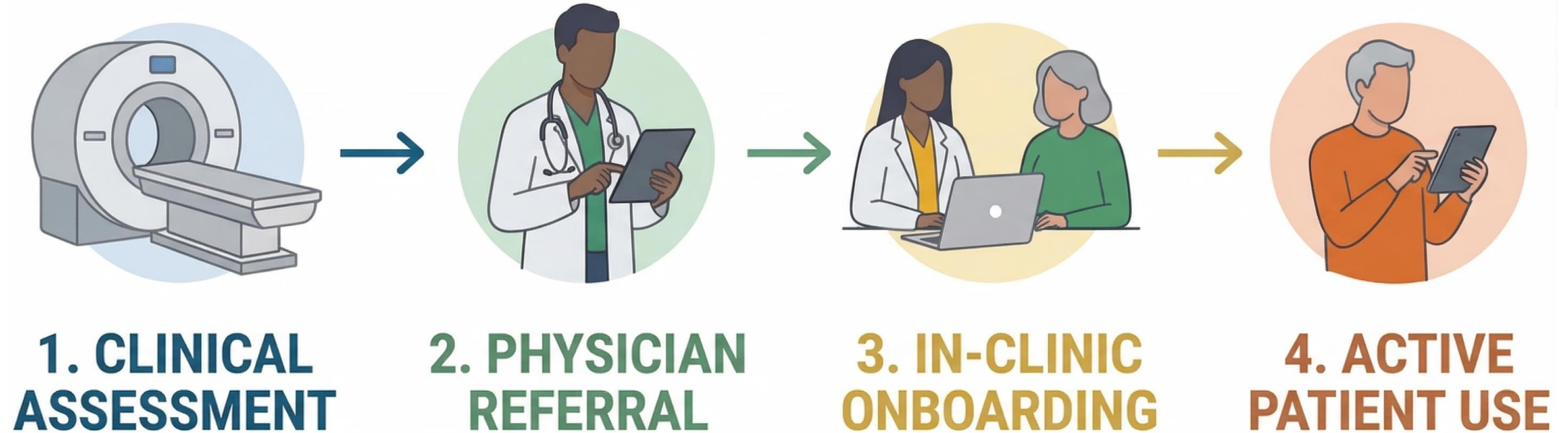


**2. PHYSICIAN
REFERRAL**

Distribution of AtaxAI to patients through healthcare providers



Distribution of AtaxAI to patients through healthcare providers



AtaxAI: 30-Month Development Timeline



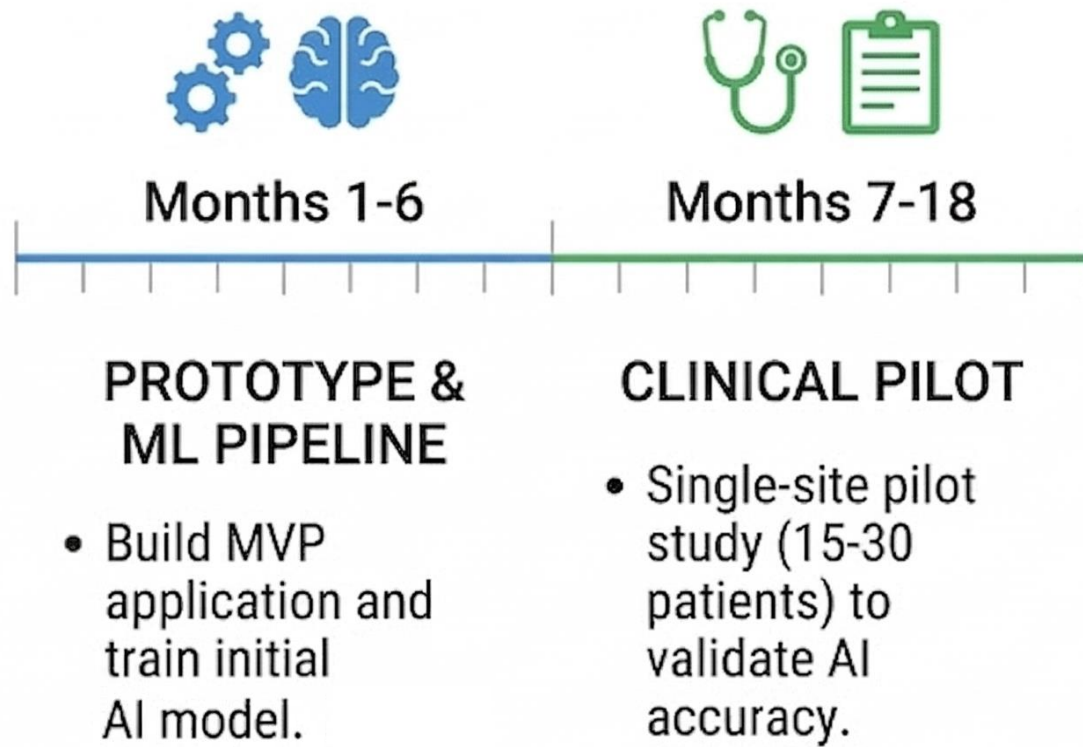
Months 1-6



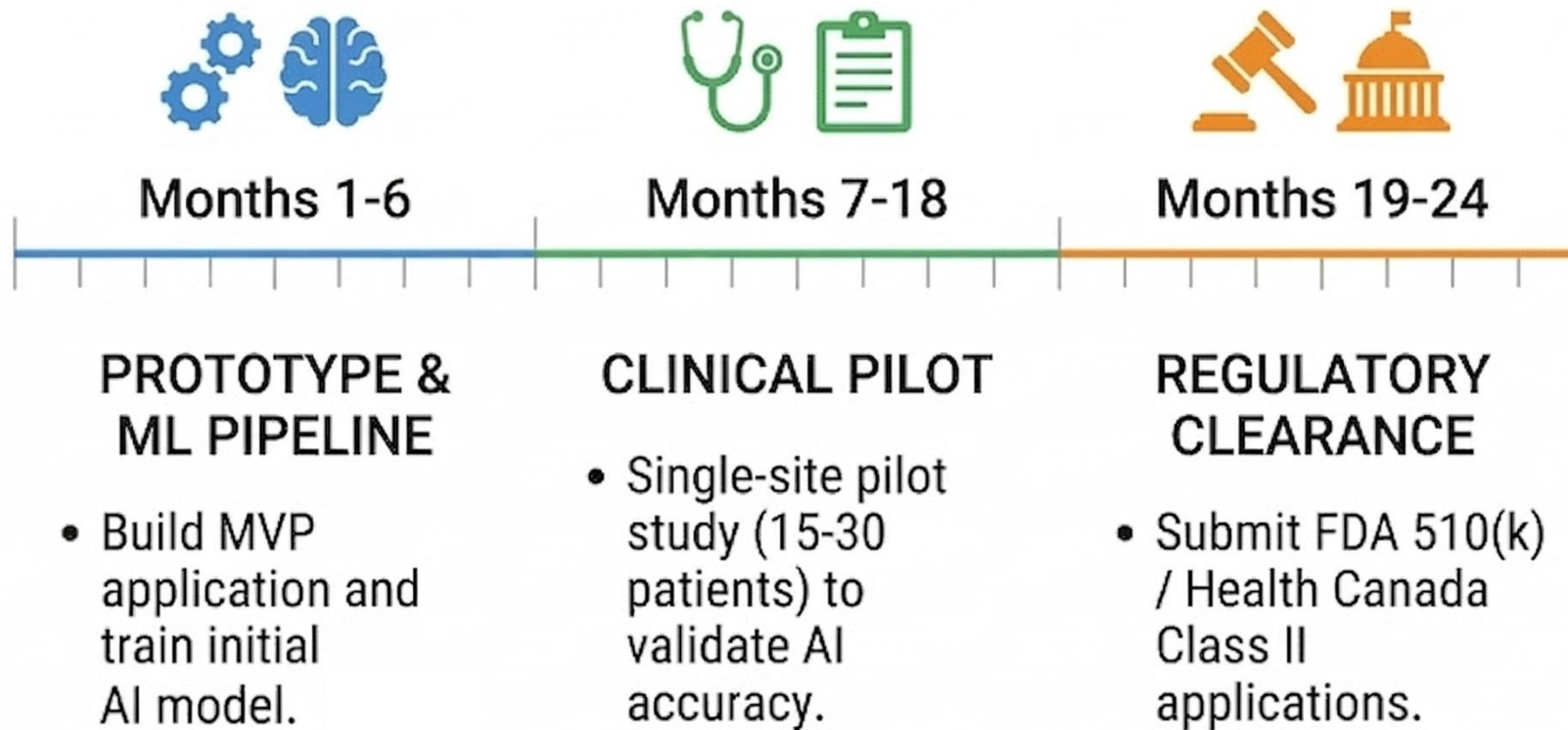
PROTOTYPE & ML PIPELINE

- Build MVP application and train initial AI model.

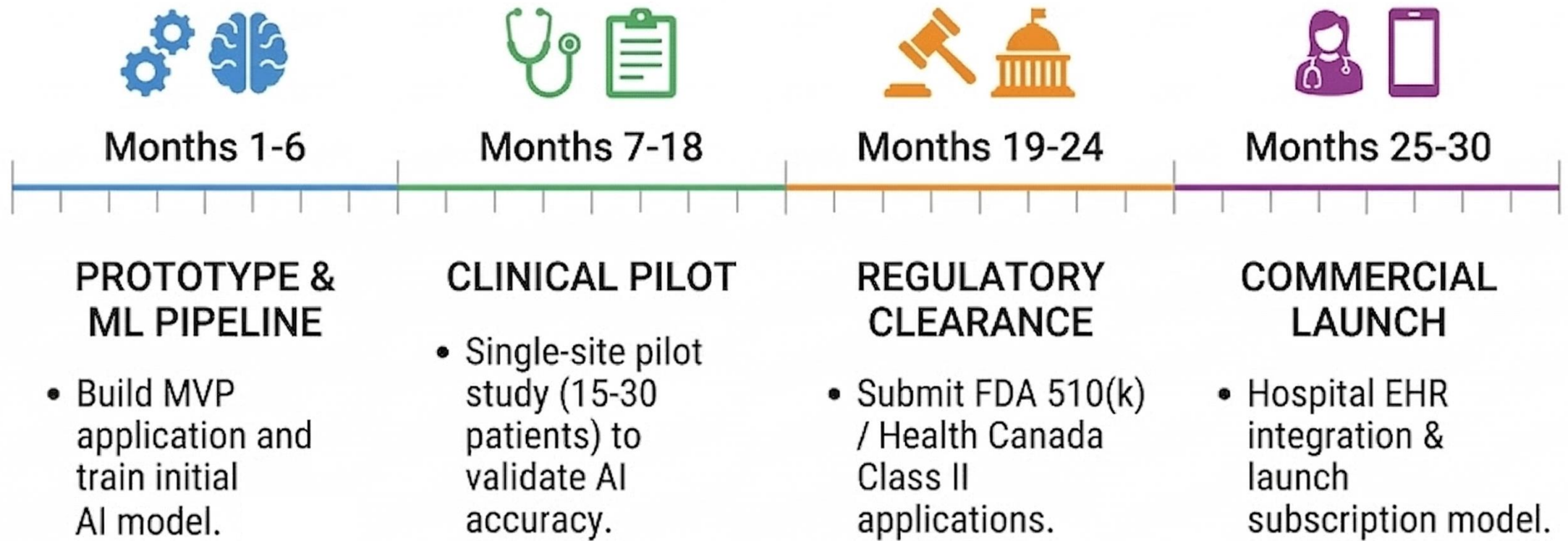
AtaxAI: 30-Month Development Timeline



AtaxAI: 30-Month Development Timeline



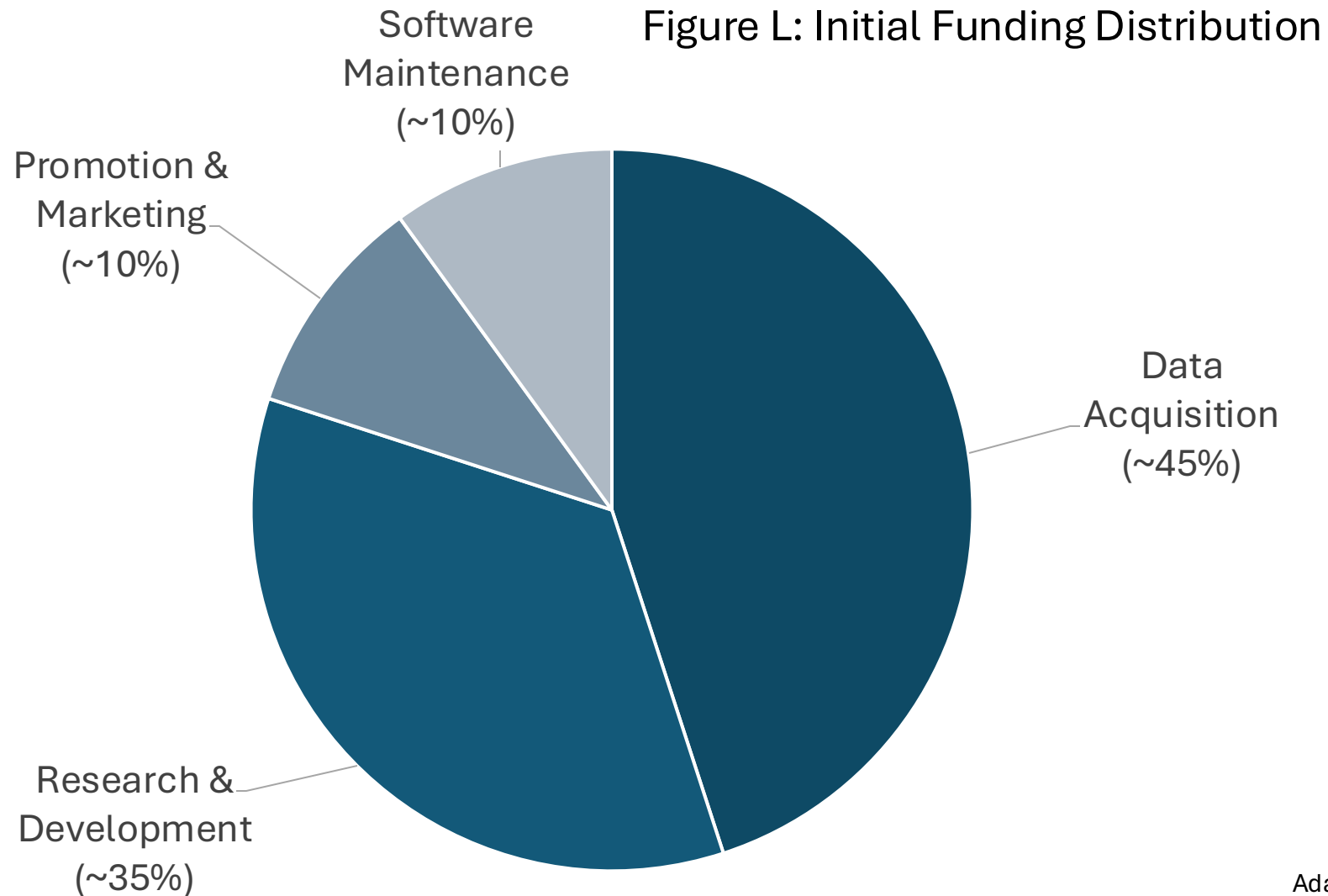
AtaxAI: 30-Month Development Timeline



Utilizing external funding, AtaxAI can position itself as a disruptive innovator in the field of neurodegenerative disease

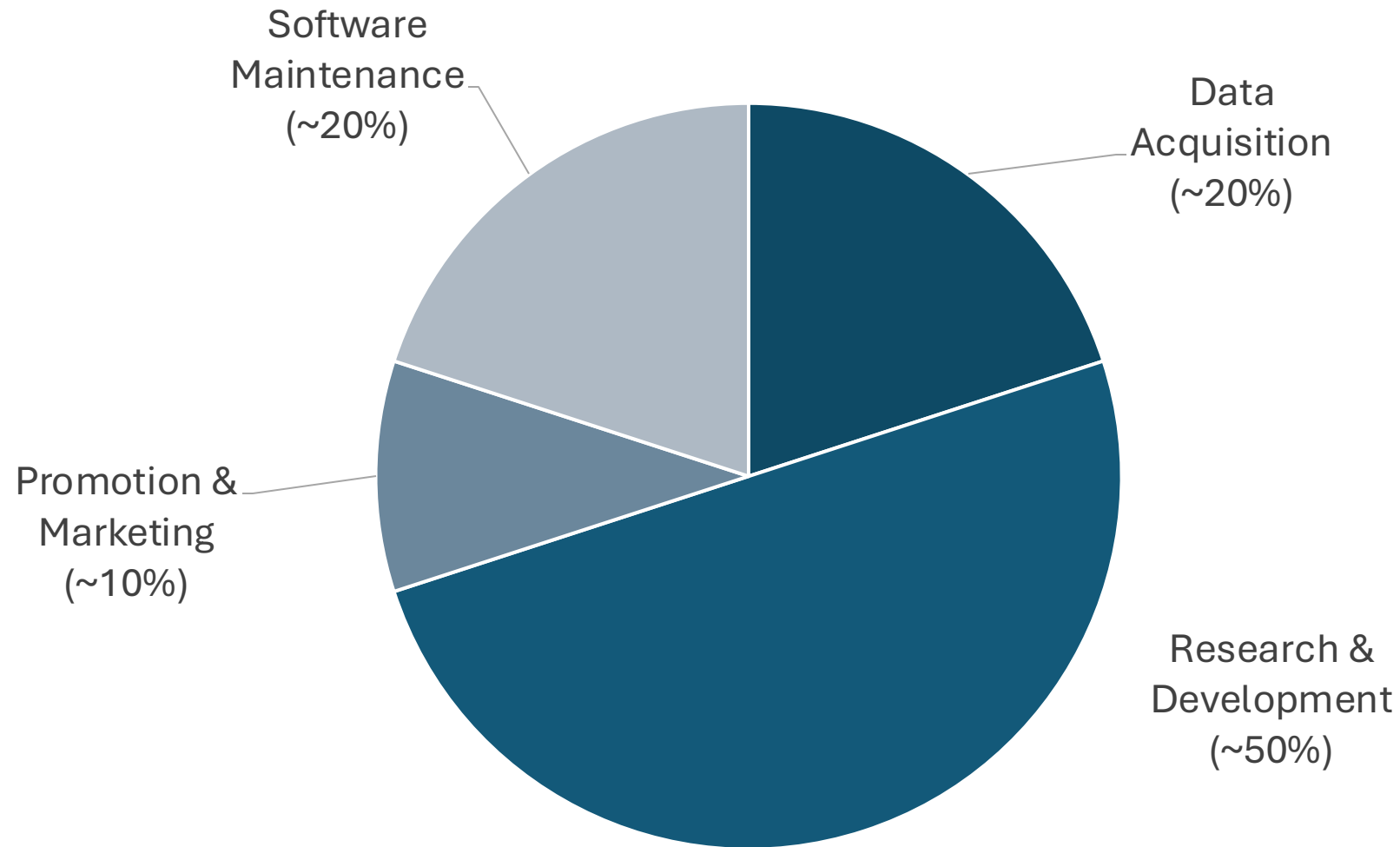


Initial funding allocation will include marketing towards clinics and hospitals, data purchases from imaging centers, and research and development



Over the long-term, funding will pivot towards research & development to investigate applications in other neurodegenerative contexts

Figure G: Long-term Funding Distribution



Thank you for
your time and
investment in
AtaxAI



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Appendix

Approximately 15000 people worldwide have FA [10], representing a possible annual revenue of approximately \$27 to \$36 million

\$150-\$200/mo subscription-
based model [11]
multiplied by the current
market of 15000

=

\$27-36 million

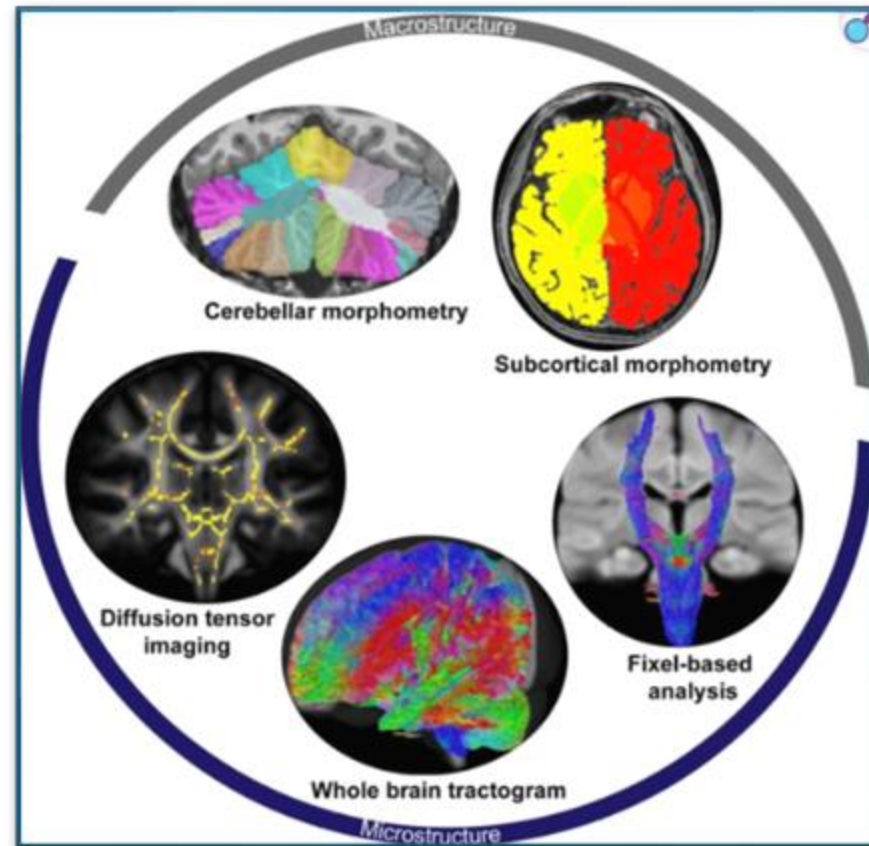
Developing AtaxAI for other
neurodegenerative diseases

\$150-\$200/mo subscription-
based model [11]
multiplied by 5% [12] of the
neurodegenerative disease market

=

\$7.2-9.6 billion

Integrating Artificial Intelligence To Build Complete Biocognitive Profiles For Patients With Friedrich's Ataxia



- Purchasing imaging and patient data from healthcare networks
- Allowing patients to self-submit their records and previous imaging
- Administering cognitive and physical self-assessments to users for benchmark parameters

Ensuring ethical AI use and data handling by integrating datasets with hospital networks



Data access through
health network EHRs [14]



Data disclosure is
voluntary and retractable



Locally-hosted AI server to
ensure data safety

Identification of Friedreich's Ataxia with MRI-based Machine Learning

